

Controller: This represents the action taken based on the sensor data, as shown in Figure 4. Through the *function* node, appropriate conditions for the oxygenator are coded using JavaScript. The oxygenator is used to increase the oxygen content in the tank, as a result of the validation of the DO content in the water found by the DO sensor. If the content drops below 5, the oxygenator is turned on to increase it. The temperature filter heats or cools the water temperature to maintain the tropical temperature in the tank as a result of the information sent to the controller. If the temperature drops or increases due to the environment around, the filter regulates and maintains the desired temperature.

Pseudocode for oxygenator (to be written in function node using JavaScript coding)

Initialize oxygeneate =0.25

Check for tank clean status

If DO <5 then add DO +oxygeneate

Send status as "Low DO level, turn on Oxygenate"

Else send status as "Normal DO level"

Tank cleaning: The tank cleaning process is an action simulated as a result of the validation of the pH value sent to the controller by the pH sensor, as seen in Figure 5. Through the *function* node, appropriate conditions for the pH value are coded using JavaScript. If the pH value drops below 6.4 or increases above 7.6, it indicates either more acidic or basic content in the water, and the cleaning process is initiated. The system then alerts other sensor modules to halt. This module has a water level indicator to find the level of the water in the tank, a water inlet, and an outlet. When the process starts, the water outlet is opened and 70 per cent of the water is let out. Then the water inlet is turned on to fill the water in the tank.

Cloud integration: The data is sent to the cloud using the *cloudant* node. Also, Node-RED is hosted in the

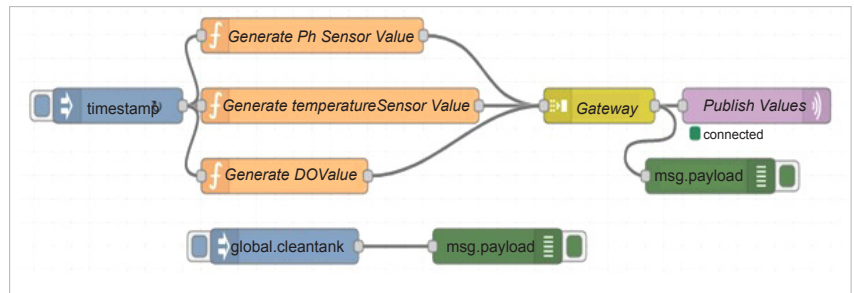


Figure 3: Sensor data generation through Node-RED

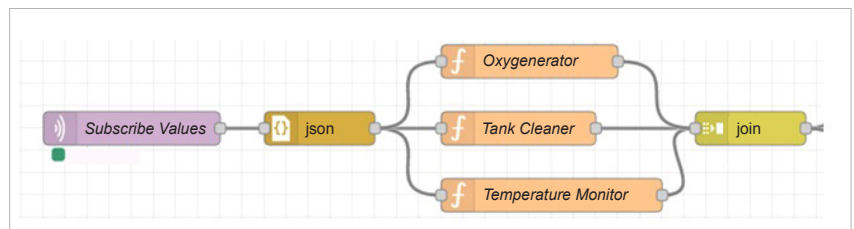


Figure 4: Oxygenator using Node-RED

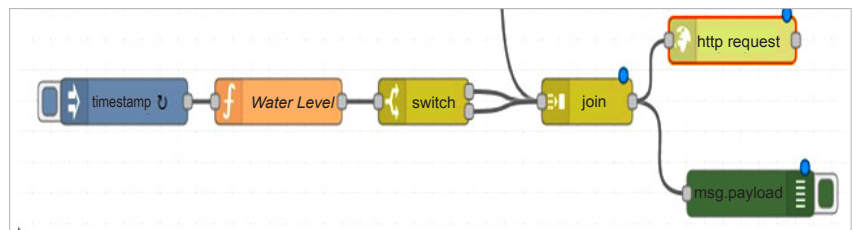


Figure 5: pH monitor and tank cleaner controller by Node-RED

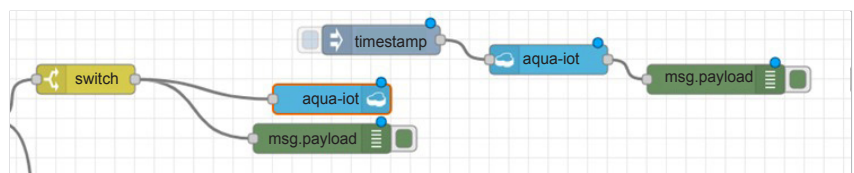


Figure 6: Cloudant out in Node-RED

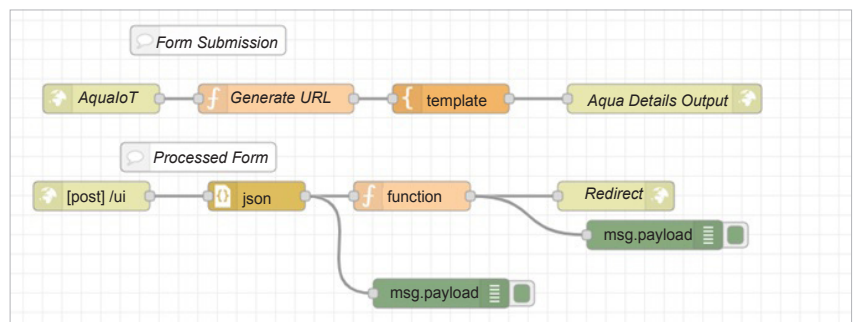


Figure 7: User interface design through Node-RED

cloud using IBM's Cloudant database, as presented in Figure 6. The output

dashboard is also hosted in the public network to allow remote access to the system using the mobile or Web.